

Pearson Correlation for Ordinal Variables: Consolidated Analysis

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1 Extremal Pearson Correlation for Ordinal Variables

1.1 1. Problem Setting and Notation

We consider two ordinal variables:

- X with K_X ordered categories $0, 1, \dots, K_X - 1$
- Y with K_Y ordered categories $0, 1, \dots, K_Y - 1$

Marginals given by:

$$p_X(i) = P(X = i), \quad p_Y(j) = P(Y = j)$$

or counts:

$$n_{X=i}, \quad n_{Y=j}, \quad N = \sum_i n_{X=i} = \sum_j n_{Y=j}$$

CDFs:

$$F_X(i) = \sum_{k \leq i} p_X(k), \quad F_Y(j) = \sum_{k \leq j} p_Y(k)$$

Pearson correlation:

$$r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad E[XY] = \sum_{i=0}^{K_X-1} \sum_{j=0}^{K_Y-1} x_i y_j P(X = x_i, Y = y_j)$$

For Spearman's correlation, replace X, Y by their ranks R_X, R_Y .

(Sources: *Proofs-OrdinalVariables.md*; *Ordinal Variables and Correlation.md*; *BFbounds.md*;
Math Proof GPT – Whitt 1976 Paper.md)

1.2 2. Fréchet–Hoeffding / Boole–Fréchet Bounds

Any joint CDF F_{XY} with marginals F_X, F_Y satisfies:

$$\max\{F_X(i) + F_Y(j) - 1, 0\} \leq F_{XY}(i, j) \leq \min\{F_X(i), F_Y(j)\}$$

(*BFbounds.md*)

- **Upper bound (comonotonic):**

$$F_{XY}^{\text{upper}}(i, j) = \min\{F_X(i), F_Y(j)\}$$

- **Lower bound (countermonotonic):**

$$F_{XY}^{\text{lower}}(i, j) = \max\{F_X(i) + F_Y(j) - 1, 0\}$$

1.3 3. Maximizing and Minimizing r

1.3.0.1 3.1 Max r — Comonotonic Coupling

Claim: The maximum correlation $r_{\max}$ is achieved by the comonotonic (FH upper) arrangement.

Proofs / Arguments: - **FH upper pmf construction:** In discrete case, differences of CDF steps allocate mass along the diagonal, pairing high X with high Y .

(*BFbounds.md*; *Proofs-OrdinalVariables.md*) - **Rearrangement inequality (ranks):** For ranks R_X, R_Y , $\sum R_X R_Y$ is maximized when sequences are ordered the same way.

(*Ordinal Variables and Correlation.md*) - **Quantile coupling:** $U \sim \text{Unif}(0, 1)$, $X = F_X^{-1}(U)$, $Y = F_Y^{-1}(U)$.

(*Math Proof GPT – Whitt 1976 Paper.md*)

1.3.0.2 3.2 Min r — Countermonotonic Coupling

Claim: The minimal correlation $r_{\min}$ corresponds conceptually to the countermonotonic (FH lower) arrangement.

Proofs / Arguments: - **FH lower pmf construction:** Differences of the lower-bound CDF.

(*BFbounds.md*) - **Opposite-order pairing:** Sort X descending, Y ascending; achieves best feasible negative correlation.

(*Proofs-OrdinalVariables.md*; *Ordinal Variables and Correlation.md*) - **Quantile coupling:** $X = F_X^{-1}(U)$, $Y = F_Y^{-1}(1 - U)$ in ideal continuous case.

(*Math Proof GPT – Whitt 1976 Paper.md*)

Note: In discrete settings, FH lower may not be attainable; see Section 5.

1.4 4. Binary (2×2) Cases and Examples

(*From Cheemera v1.1 – Pearson Correlation Analysis Implications.md*)

1.4.0.1 4.1 Balanced marginals (70–30, 70–30)

- Perfect diagonal fill yields $r_{\max} = 1$.

1.4.0.2 4.2 Skew vs balanced (90–10, 50–50)

- Perfect alignment impossible; $r_{\max} < 1$ due to capacity limits.

1.5 5. Attainability and Symmetry (Updated)

1.5.1 5.1 Maximal correlation

- **Result:** In the discrete ordinal setting, $r_{\max}$ is **always** attained by the FH upper (comonotonic) construction.

(*BFbounds.md*; *Proofs-OrdinalVariables.md*; *Ordinal Variables and Correlation.md*)